

Achieving professional qualifications using micro-credentials: a case of small packages and big challenges in higher education

Rohit Raj

Department of Information Management, Chaoyang University of Technology, Taichung, Taiwan

Arpit Singh

Department of Information Systems and Analytics, Jindal Global Business School, OP Jindal Global University, Sonapat, India

Vimal Kumar

Department of Information Management, Chaoyang University of Technology, Taichung, Taiwan, and

Pratima Verma

Department of Strategic Management, Indian Institute of Management Kozhikode, Kozhikode, India

Abstract

Purpose – This study examined the factors impeding the implementation of micro-credentials and accepting it as a credible source of earning professional qualifications and certifications necessary for pursuing higher education or other career goals.

Design/methodology/approach – The factors were identified by reflecting on the recent literature and Internet resources coupled with in-depth brainstorming with experts in the field of micro-credentials including educators, learners and employers. Two ranking methods, namely Preference Ranking for Organization Method for Enrichment Evaluation (PROMETHEE) and multi-objective optimization based on ratio analysis (MOORA), are used together to rank the major challenges.

Findings – The results of this study present that lack of clear definitions, ambiguous course descriptions, lack of accreditation and quality assurance, unclear remuneration policies, lack of coordination between learning hours and learning outcomes, the inadequate volume of learning, and lack of acceptance by individuals and organizations are the top-ranked and the most significant barriers in the implementation of micro-credentials.

Research limitations/implications – The findings can be used by educational institutions, organizations and policymakers to better understand the issues and develop strategies to address them, making micro-credentials a more recognized form of education and qualifications.



This paper forms part of a special section “Micro-credential as a Digital Enabler for Higher Education Ecosystems”, guest edited by Muhammad Ali, Syed Ali Raza, Chin Hong Puah and Wasim Qazi.

The authors would like to thank the two anonymous reviewers, Associate Editor, and Editor-in-Chief for their valuable comments and suggestions that helped to improve the manuscript.

Funding: The authors received no financial support for the research, authorship and/or publication of this article.

Declaration of conflicting interests: The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

Originality/value – The novelty of this study is to identify the primary factors influencing the implementation of micro-credentials from the educators', students' and employers' perspectives and to prioritize those using ranking methods such as PROMETHEE and MOORA.

Keywords Micro-credentials, Online certifications, Implementation, Barriers

Paper type Research paper

1. Introduction

The term “micro-credential” first emerged around 2013, but it existed conceptually in the form of offering credentials for a much longer period (Brown *et al.*, 2021). For instance, software companies have provided certification of competence in standardized processes such as program management and quality assurance. In the wake of a pandemic, the global workforce and economy are at a critical juncture where organizations are confronted with new economic and workforce challenges (Gigauri, 2021; Paliwal *et al.*, 2010; Raj *et al.*, 2023b; Rajak *et al.*, 2023). In particular, the increasing expense of higher education, employer's focus on essential abilities and expertise, and students' difficulties in securing good job opportunities have prompted universities, independent certifying bodies and creators of national qualification reference frameworks to rethink the entire credentials spectrum (Gordon *et al.*, 2021). The economic shifts and rethinking of micro-credentialing had already been emerging due to high employer demands for qualified personnel, good employment opportunities, career advancements and mobility in career paths (McGreal *et al.*, 2022). There have been multiple reports from countries around the world regarding employment and skilled workforce shortages. According to the National Skills Commission, Australia's skill shortage has nearly doubled from 2021 to 2022 (Dawson, 2021). Less than 45% of Irish employers have agreed to possess the talent and skills needed to achieve quality employment (Murphy and O'Sullivan, 2021). As per Equinix, 2022 Global Tech Trends Survey, 74% of IT decision-makers in India perceive a shortage of personnel with IT skills as one of the major impediments to the growth of the IT sector in India (Curtis *et al.*, 2022). A staggering 40% of European employers have experienced difficulties finding employees with the required skills (Forbes, 2023). Beyond the traditional credentials base of degrees and certificates, the boundaries of validated formal and non-formal learning are expanding (Brunello and Wruuck, 2021). This is leading to an approach in which all pertinent skill and competency learning can be validated, certified and transcribed by a third party in addition to a university as part of students' aggregated learning, skills, competencies and experience. The movement toward micro-credentials, also known as alternative digital credentials (ADCs), nano degrees, or badges, has accelerated as a result of this shift in practices toward unbundling higher education (Wolz *et al.*, 2021). One research estimates that there will be at least 1,500 micro-credentials available in the middle of 2021, most of which will be based on MOOCs (McGreal and Olcott, 2022). The majority of micro-credentials—more than 75%—are in business or technology (Olcott, 2022).

The micro-credentialing is a pivotal process that equips teachers and academicians in developing essential skills and techniques that are helpful to their professional growth and subsequently to the students (Selvaratnam and Sankey, 2021). National and regional education agencies posit that every student deserves excellent teachers and building leaders who promote holistic professional and personal efficiency of the student body (Jääskelä *et al.*, 2021). According to the website of the European Commission (European Education Area, 2023) describing the use of micro-credentialing, it promotes instructors to gain knowledge and abilities to comprehend the appropriate applications that improve students' learning (Rieder, 2022). Teachers and academicians worldwide have expressed keen interest in undergoing teacher development programs that are relevant to the work they are doing in classrooms and are long-lasting rather than one-shot and sit-and-get programs (Hu *et al.*, 2021). Some universities and schools across the globe have started investigating and experimenting with “micro-credentialing” as a viable opportunity to impart high-quality, relevant, flexible and research-driven professional development experience for teachers and academicians worldwide

(Boud and Jorre de St Jorre, 2021). Micro-credentials offer validation to the learning acquired by teachers and receive recognition on a global platform that further assists in achieving milestones in professional learning trajectories (Martinez-Marroquin and Male, 2021). The reason why micro-credentialing appeals to the academic community is the concept of breaking down complex instructional skills into fundamental parts (West and Cheng, 2022). Educators can gain competence in smaller fundamental parts and gradually acquire expertise in an entire subject to demonstrate mastery of complex skills. Teachers therefore are enabled to focus only on the primary skills that are relevant to the courses they are undertaking or any other academic activity and are saved from the unnecessary efforts required to claim competence in an entire discipline that is of no immediate importance to them (Kohler *et al.*, 2021). Micro-credentialing offers multiple noteworthy benefits to the government at national and state levels. Firstly, the government can streamline micro-credentialing initiatives that are aligned with the teaching standards and requirements (Brown *et al.*, 2021). For instance, adequate and proper support should be provided to teachers to teach specific language learners. Secondly, the mode of offering micro-credential training is online which makes it accessible to educators who are placed in remote locations and with different geographical locations with different working hours (McGreal *et al.*, 2022; Kumar *et al.*, 2022b). Lastly, micro-credentialing assists educators in understanding their own professional needs (White, 2021). They may be viewed by students as more approachable than a general degree. Learning may also be tailored to a student's requirements or, for seasoned employees, utilized to upgrade their skills in a particular field. A set of "stackable" micro-credentials that may be completed over time may also be interesting to pursue to progressively aggregate the learning that may be comparable to (and recognized as) a regular certification. Micro-credentials might be utilized to evaluate and acknowledge past education and expertise.

The discussion on micro-credentialing revolves majorly around three concurrent factors. Firstly, micro-credentials enable employers to access the relevant information on candidates about their qualifications and skills they have acquired through both formal and informal learning over time. Employers can benefit the most due to the detailed and granular nature of the details on the skills through micro-credentials. This information can be useful in making recruitment and promoting decisions, deciding the appraisals and reward points, and identifying gaps in employee skills and knowledge which can help authorities in designing personnel development programs. The second factor necessitating the implementation of micro-credentials is the desire of individuals to have their knowledge and learning recognized in a timely fashion that is acceptable on global standards as opposed to macro-credentialing where it takes a huge amount of resources and time and is relatively expensive. The ability of present and developing technology to satisfy these demands of both companies and individuals is the third contributing element (Huynh-Cam *et al.*, 2022). Together, these elements offer a chance for a model that shifts the credentialing approach from one that focuses on individual acquisition and maintenance to one that focuses on institution-focused delivery and record-keeping.

The main goal of this research is to provide a briefing paper outlining the major methodological and practical challenges associated with recognizing micro-credentials for both academic and employment reasons. Based on a series of in-depth interviews with important participants (students, institutions, governments and companies), it identifies the major hurdles in embracing micro-credentialing across universities and organizations.

This gives the educators and employers the opportunity to either:

- (1) Offer better explanations on how to utilize those instruments in the context of providing micro-credentials; or
- (2) Recommend optimizations to the instruments enabling the supply of micro-credentials.

2. Literature review

2.1 Origin of micro-credentials

Micro-credentials are rooted theoretically in the “digital badge” movement that was geared to provide adult learning to the workforce (Ifenthaler *et al.*, 2016). Open badges, created by the Mozilla Foundation in 2013, was the pioneer in this movement with ample funding from the MacArthur Foundation (Frederiksen, 2013). Open badges provided a platform to individuals who wished to gain badges that meet certain technical standards validating the skills and learning acquired on a lifelong basis (MacKinnon, 2021). One of the first revolutionary efforts in the domain of micro-credentialing was a working paper authored by the Mozilla Foundation in collaboration with Peer 2 Peer University that advocated the vision of digital badges as a lifelong learning vehicle for the working community. This was later reinforced by the Clinton Global Initiative to expand access to digital badges to assist workers in certifying and acquiring skills to improve their social and economic futures (Aydarova, 2022). This led to multiple issuers creating and posting badges on multiple sites across the web that can be earned in a multitude of disciplines and fields. In the year 2017, the IMS consortium took over a substantial portion of Open Badges’ responsibilities. The importance of digital badges in validating functional skills at the workplace was the focal point of the research conducted by the American Institutes for Research in 2013 (Finkelstein *et al.*, 2013). Several industries have started using digital badges to demonstrate and recognize specific competencies. For instance, the Interstate Renewable Energy Council awards teachers and trainers with badges that acknowledge their capacity to guide learning (Charania, 2022). Those who exhibit proficiency in technical writing are given badges by the organization such as TechWriter Certification (De Monte, 2017). Education groups have also seen that teachers are becoming more interested in using technology to interact with their colleagues and access improved professional learning opportunities, such as Twitter conversations, online courses and EdCamps (Barry, 2022; Singh *et al.*, 2023). The investigation of micro-credentials for teacher professional development probably benefited from the industry’s and adult learning advocates’ increased interest in and experimentation with digital badges.

2.2 Micro-credentials: educators’, students’ and employers’ perspectives

Micro-credentials are efficient tools for educators to develop and acquire competencies in new skills useful for professional development. External agents can provide feedback on the skills earned by the credentials which can be used to gain recognition in academia (Hunt *et al.*, 2020). Each micro-credential typically tackles a narrow range of distinct educational activities. There are several distinct “issuers” who have produced hundreds of different micro-credentials. Some of them, such as those on the public pages of the Digital Promise website, are open to anybody who wants to try to earn it (Ghasia *et al.*, 2019). Others are not accessible to the general public because they were created expressly for a state, district, or program. Depending on the agreements made between states and issuers, the issuer may charge a different amount, although normally the cost of scoring is included (Randall and West, 2022). Micro-credentials form an integral part of a stack of specialized areas of teaching. For instance, the course stack “Learning coding” can be divided into several micro-credentials including “learning coding fundamentals”, “coding early objects and algorithms”, and “computer programming linear and non-linear data structures” (Selvaratnam and Sankey, 2021).

Things in the education sector have a propensity to change quickly, giving the impression that everything is happening suddenly. Today’s students are really worried about remaining “future-proof,” or being able to keep up with such developments in order to remain competitive when they join the job (Wheelahan and Moodie, 2021). A “micro-credential” must be earned differently as compared to a “macro-credential” (Boud and Jorre de St Jorre, 2021). First of all, students are given the freedom to select and ignore the specific lessons they wish to study. They receive the prize as soon as they’ve demonstrated that they know enough

(Ruddy and Ponte, 2019). These prizes typically take the shape of digital badges that students may display on their online profiles and resumes to potential employers (LaMagna, 2017). Micro-credentials are more affordable than regular certificates, and students may do them as quickly or slowly as they can (Lim *et al.*, 2018). When micro-credentials started gaining momentum, it was not the traditional educational institutions that made them known instead it was the non-traditional educational platforms such as Udacity's "nano-degree", edX's "Micromasters" and Coursera's "MasterTrack" certificates that brought micro-credentials to the forefront. Micro-credentials are useful at any point in one's career (Kato *et al.*, 2020).

Once employed, one may continue to be future-proof by constantly upgrading one's skills by obtaining new digital badges or degrees. Upskilling used to frequently include enrolling in a part-time course, which would disrupt one's life and employment. Working individuals might continue to do their occupations and continue to learn anytime they wish by using micro-credentials (Milligan and Kennedy, 2017). To comprehend and meet the demands of working adults, micro-credential providers frequently cooperate with experts in the sector (Rottmann and Duggan, 2021). For instance, Udacity collaborates closely with organizations like Google, AT&T, Salesforce and even Facebook to make sure that their courses meet the needs of working adults. Concerns about having the necessary information and abilities to be competitive and remain employed are common among students and workers. They can easily and swiftly get what they need with micro-credentials. The inexpensive cost of the courses is made even more apparent by the fact that students are permitted to utilize free educational resources like e-books for studying and plagiarism detection software for any written homework (Milligan and Kennedy, 2017). Having micro-credentials might be helpful for many students looking for competitive internships. Two recent graduates from the same university may have similar career profiles. However, if one has the appropriate digital badges, she/he will be able to stand out more and convince companies that they are the superior applicant. The major hurdles in implementing micro-credentials are collected in Table 1 following an extensive literature review and consultation with the key stakeholders in micro-credentialing.

2.3 Major hurdles in the implementation of micro-credentials

2.3.1 Program outline (PO). An in-depth description of the components, their relationships and the steps the program will take to accomplish its objectives is usually provided in a program outline, which is a high-level explanation of the framework and process of any microcredit (Olcott, 2022). It is essential to have support and advocacy from a variety of stakeholders, including educational institutions, employers, industry associations and regulatory bodies, for micro-credentials to become widely accepted and recognized (Varadarajan *et al.*, 2023). Without a concerted effort, awareness and buy-in may be low, making it challenging to build a strong ecosystem for micro-credentials. To develop a clear understanding of the value and function of micro-credentials, stakeholders must cooperate and coordinate (McGreal *et al.*, 2022). For micro-credentials to be used effectively, they must have a precise and simple meaning. If the idea and intent of micro-credentials are not made clear, stakeholders may become perplexed and misunderstand them. Different perspectives on what qualifies as a micro-credential may obstruct the creation of uniform standards, quality control procedures and recognition frameworks. The significance and relevance of micro-credentials may not be understood by learners, which would hinder their uptake (Olcott, 2022). Micro-credentials frequently make up a small portion of a whole certification or degree program and typically concentrate on certain skills or knowledge. However, it can be difficult when there is not a recognized amount of learning connected to micro-credentials (Boud and Jorre de St Jorre, 2021). Traditional academic certifications and degrees have set credit systems and specified completion times. Micro-credentials, on the other hand, might not have standardized measures for evaluating the breadth and depth of learning results.

Table 1.
Major hurdles in
implementing micro-
credentials

Variables	Criteria	References
Programme outline (PO)	PO1 The absence of a strong, unified stakeholder push for micro-credentials	Tinsley <i>et al.</i> (2022), Flintberg (2022), Resei <i>et al.</i> (2018)
	PO2 Absence of a clear definition that can be easily understood without any specific expertise or discipline background	Cook (2021), Del Toro (2021), Lexman <i>et al.</i> (2020)
	PO3 Absence of a recognized volume of learning	Kumar <i>et al.</i> (2022a), McGreal <i>et al.</i> (2022), Rosenbaum <i>et al.</i> (2021)
	PO4 Pre-designed learning curricula that are difficult to divide into smaller sections	Quigley <i>et al.</i> (2022), Clausen (2022), Mardisentosa <i>et al.</i> (2021)
Programme acceptance (PA)	PA1 Lack of acceptance by one professional or occupational group across multiple jurisdictions	Ward <i>et al.</i> (2021), McGreal <i>et al.</i> (2022), Wheelahan and Moodie (2021)
	PA2 Lack of acceptance by employers	Ghasia <i>et al.</i> (2019), Tereseviciene <i>et al.</i> (2020), Kato <i>et al.</i> (2020)
	PA3 Lack of acceptance by individuals	Selvaratnam and Sankey (2021), McGreal <i>et al.</i> (2022), Wheelahan and Moodie (2021)
Programme end-results (PE)	PE1 The mismatch between learning hours and results	Wheelahan and Moodie (2021), Cook (2021), AlHamad and AlAmri (2021)
	PE2 Lack of detailed identity verification measures	Mardisentosa <i>et al.</i> (2021), Rossiter and Tynan (2019), Kato <i>et al.</i> (2020)
	PE3 Lack of detailed process assessment methods	McGreal <i>et al.</i> (2022), Boud and Jorre de St Jorre (2021), Rossiter and Tynan (2019), Hunt <i>et al.</i> (2020)
	PE4 Course descriptions that differ, particularly when every comparable course is compared, such as learning outcomes	Bigelow <i>et al.</i> (2022), Jamaluddin <i>et al.</i> (2021), Thi Ngoc Ha <i>et al.</i> (2022)
Programme regulation (PR)	PR1 Unclear remuneration policies	McCartney and Rick (2021), Rosenbaum <i>et al.</i> (2021), Thi Ngoc Ha <i>et al.</i> (2022)
	PR2 Lack of trust between different institutions	McGreal <i>et al.</i> (2022), Selvaratnam and Sankey (2021), Tereseviciene <i>et al.</i> (2020)
	PR3 The problem of accreditation and quality assurance is the problem of security issues of credentials	McGreal <i>et al.</i> (2022), Brown <i>et al.</i> (2021), Msweli <i>et al.</i> (2022)

Source(s): Authors' own

This could impede the acceptance and transferability of different micro-credentials by causing doubt about the degree of skill and comparability among them (Mashford-Pringle *et al.*, 2023). Pre-designed learning curricula sometimes have a rigid structure that is difficult to break down into smaller sections or modules, especially when they were created for traditional degree programs or longer-duration courses. These curricula are frequently created to provide a thorough understanding of a subject over a long period, making it difficult to divide them into smaller, independent courses that are consistent with the idea of micro-credentials (Ghasia *et al.*, 2019). Not modular enough Micro-credentials are frequently created to be modular, allowing students to concentrate on particular knowledge or skill areas. The inability to detach and isolate smaller pieces without interfering with the overall coherence and continuity of the learning experience occurs when a pre-designed learning curriculum lacks modularity. This may make it more difficult to develop micro-credentials based on current curriculums.

2.3.2 Program acceptance (PA). Micro-credential uptake and use may be hampered if a particular professional or occupational organization does not acknowledge or appreciate them. For admittance or progress, professions frequently have set norms and qualifications, which may not include acceptance of micro-credentials (Zain, 2023). Micro-credential holders may have fewer opportunities to be respected and regarded in their industries as a result of this lack of acceptance. It can also result in disparities and inconsistencies in how micro-credentials are viewed and accepted across various jurisdictions, making it challenging for people to advance or be recognized outside of a limited geographic area (Fogarty et al., 2023). The recognition and acceptance of micro-credentials by employers are crucial for establishing their meaningful impact and value in the job market (Garg et al., 2023). If employers are uncertain about the utility of micro-credentials in estimating a candidate's aptitude they are less likely to consider them for the hiring process and career advancement opportunities. This can cause a mismatch between the skills and knowledge individuals acquire through micro-credentials and the expectations and requirements of employers (Zou et al., 2023). Without employer acceptance, individuals may not perceive micro-credentials as viable investments of their time and efforts thereby undermining their willingness to pursue them. Individuals themselves play a pivotal role in the successful implementation of micro-credentialing. If they do not perceive micro-credentials as a value-adding component relevant to their career goals, they are less likely to pursue them. Individual acceptance may be hampered by elements including lack of knowledge, skepticism about the validity of micro-credentials, or a desire for more conventional educational and credentialing methods. The market for micro-credentials may stay low without individual buy-in and involvement, limiting their influence and possibility for wide-scale adoption.

2.3.3 Program end-results (PE). Micro-credentials are designed to provide focused and specific learning outcomes in a shorter timeframe compared to traditional education programs. However, a micro-credential's credibility and perceived value may be damaged if there is a considerable discrepancy between the intended learning outcomes and the allotted learning hours (Duklas, 2020). If the expected results do not match the time and effort put in, learners and stakeholders may doubt the depth and rigor of the learning process (Yanek, 2020). Micro-credentialing can be deeply affected by the lack of identity verification methods for individuals participating in it (Kumar et al., 2022a). There is a high possibility of impersonators and fraudulent elements causing disruptions in the process due to the lack of proper identification methods. This can seriously hamper the credibility and trustworthiness of micro-credentials (Carmichael and Eaton, 2023). Evaluation of the learning process and results is a crucial component of micro-credentialing. The evaluation criteria may differ and be inconsistent if there aren't detailed and standardized ways for evaluating the micro-credentialing processes and activities. This may have an impact on the micro-credentials validity and comparability among various people or organizations (Navanitha et al., 2022). It is pivotal to ensure that the course descriptions, learning outcomes and competencies acquired are accurately mentioned in the course descriptions when evaluating micro-credentials. This is essential to maintain consistency across course descriptions, particularly when comparing comparable courses. Higher education must abandon conventional methods and adopt a customized, industry-focused strategy to close the skills gap between industry needs and student abilities. According to competency ratings, the research suggests employing learning analytics to recommend personalized micro-credentials (Messaoud et al., 2022). The major problem studied in the research by Mardisentosa et al. (2021) is the need for assessment and management systems that ensure the confidentiality, permanence and transparency of student documents in micro-credentialing using blockchain technology. By implementing such systems, students can personalize their education, create permanent documentation and determine their lifelong learning path. The research focuses on analyzing the use of blockchain-based micro-credentials for student achievement records in universities, highlighting the benefits of durability, security and streamlined administration (Mardisentosa et al., 2021).

2.3.4 Program regulation (PR). Micro-credential deployment may be hampered by ambiguous compensation regulations. Micro-credentials' value is hampered by the absence of clear standards or frameworks for identifying and honoring those who earn them. Without obvious incentives, people could be less eager to pursue micro-credentials and organizations might be less likely to provide them. The research by [Aydarova \(2021\)](#) pointed out the major steps that can be taken to facilitate the implementation of micro-credentials for teachers' professional growth and career progression. It draws attention to the various strategies used by nations to select and encourage micro-credential membership. To prevent gaming and assist failing schools, the paper advises developing clear frameworks, connecting micro-credentials with current systems, integrating LEAs and school leaders, offering incentives, guaranteeing portability and encouraging equal resource distribution ([Aydarova, 2021](#)). The use of micro-credentials may be hindered by a lack of trust between various organizations ([Pirkkalainen et al., 2023](#)). The credibility and acceptance of micro-credentials depend on cooperation and acknowledgment between educational institutions and industrial stakeholders ([McGreal et al., 2022](#)). Institutions could be hesitant to recognize and authenticate micro-credentials from other sources, restricting their transferability and usefulness, if there is not trust and consensus on standards ([McGreal et al., 2022](#); [Pirkkalainen et al., 2023](#)). There is a big issue with accreditation and quality assurance, particularly the protection of credentials ([Varadarajan et al., 2023](#); [Ahmat et al., 2021](#)). There are questions regarding the dependability and quality of micro-credentials because they cannot always be submitted to the same exacting accreditation procedures as conventional educational programs ([Fedock et al., 2016](#)). A vital factor to take into account is the security of digital credentials. To avoid fraud and uphold belief in their worth, micro-credentials integrity and validity must be guaranteed.

3. Methodology

Discovering hurdles to the adoption of micro-credentialing across institutions and organizations is one of the most important decision-making concerns in the field of micro-credentialing. The institution and organization must recognize the effectiveness of their microcredit program, which necessitates a reassessment of several potential barriers based on the various factors listed in [Table 1](#).

The Preference Ranking Organization Method for Enrichment Evaluation (PROMETHEE) and multi-objective optimization based on ratio analysis (MOORA) methods are combined for this study. In this study, the PROMETHEE and MOORA approaches are combined since they both offer straightforward and efficient ways to rank variables and criteria. The combined method has advantages because it takes into account pairwise comparisons of the criteria as well as how each hurdle performs on each criterion. The PROMETHEE and MOORA techniques are integrated into this investigation. [Figure 1](#) illustrates how a research design framework was used to generate the selection of hurdles for the effectiveness of micro-credentialing performance in institutions and organizations. The next section has a thorough explanation of each approach and the steps involved.

3.1 PROMETHEE method

An efficient decision-making tool, PROMETHEE, provides a rating order of the major alternatives ([Brans et al., 1986](#)). It is a way of ranking solutions designed to increase competence in evaluating the criteria and identifying the best answer. This process involves the following steps ([Douplos and Zopounidis, 2004](#); [Wang et al., 2008](#); [Athawale and Chakraborty, 2010](#); [Verma et al., 2023](#)):

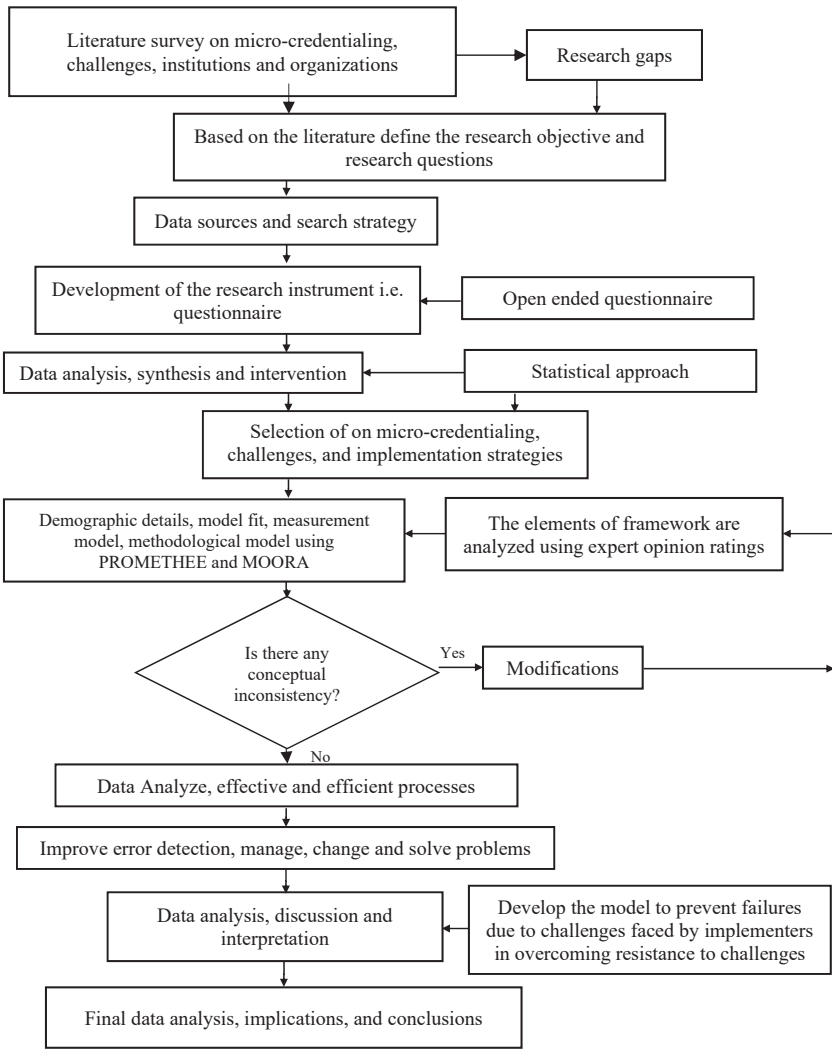


Figure 1.
Research design
framework for
the study

Source(s): Authors' own

Step 1: Utilize the following equation to normalize the decision matrix:

$$R_{i,j} = \frac{[(x_{ij}) - (x_{ij})_{\min}]}{[(x_{ij})_{\max} - (x_{ij})_{\min}]} \quad (i=1,2,3,\dots,n; j=1,2,3,\dots,m) \tag{1}$$

where X_{ij} is the performance measure of i th alternative concerning j th criterion.

Equation (1) can be revised to include non-beneficial conditions as follows:

$$R_{ij} = \frac{[(x_{ij})_{\max} - (x_{ij})]}{[(x_{ij})_{\max} - (x_{ij})_{\min}]} \quad (2)$$

Step 2: Analyze the evaluative distinctions between the i th and the other alternatives. Estimating the differences in criterion values between various alternatives pairwise is what this phase entails.

Step 3: $P_j(i, i')$, the preference function, is evaluated. In real-time applications, it may be challenging for the decision-maker to pinpoint which specific type of preference function is appropriate for each criterion and to ascertain the parameters involved due to the six types of preference functions that are currently available (Hajkowicz and Higgins, 2008). To get around this issue, the following streamlined preference function is used here:

$$P_j(i, i') = 0 \text{ if } R_{ij} \leq R_{i'j} \quad (3)$$

$$P_j(i, i') = (R_{ij} - R_{i'j}) \text{ if } R_{ij} > R_{i'j} \quad (4)$$

Step 4: Calculate the aggregate preference function taking the weights of the criterion into account.

$$\text{Aggregated preference function, } \pi(i, i') = \frac{\sum_{j=1}^m (w_j) P_j(i, i')}{\sum_{j=1}^m (w_j)} \quad (5)$$

where w_j = Relative importance (weight) of the j th criterion.

Step 5: The exiting and entering outranking fluxes should be calculated as follows:

$$\text{Leaving (or positive) flow for } i^{\text{th}} \text{ alternative, } \phi^+(i) = \frac{1}{n-1} \sum_{i'=1}^n \pi(i, i') (i \neq i') \quad (6)$$

$$\text{Entering (or negative) flow for } i^{\text{th}} \text{ alternative, } \phi^-(i) = \frac{1}{n-1} \sum_{i'=1}^n \pi(i', i) (i \neq i') \quad (7)$$

Here n denotes the number of alternatives.

Each alternative in this case has to compete with $(n-1)$ additional alternatives. The entering (negative) flux demonstrates how much one alternative is led by the other, whereas the leaving (positive) flux demonstrates how much one alternative drives the other k .

Step 6: Calculate each item's net outranking flow.

$$\phi(i) = \phi^+(i) - \phi^-(i) \quad (8)$$

Step 7: Using the $\phi(i)$ value of each option, determine the final ranking. The options with the highest $\phi(i)$ values are regarded as the best options available.

The PROMETHEE method is used to assess the dimensions of the variables. This method computes alternatives over evaluation criteria using a linguistic code or scale with detailed

information. Following the creation of the normalized decision matrix, pair-wise comparison is used to determine the preference functions for each pair of choices to aggregate them. Additionally, the overall outranking flow values for various dimension alternatives are calculated, as well as the leaving and entering fluxes for data quality dimensions.

3.2 MOORA method

A multi-criteria decision-making process called MOORA has a strong potential for thorough examination of alternatives when faced with a wide variety and multiplicity of influential factors. Brauers and Zavadskas (2006) proposed the MOORA method as one of the multi-objective optimization techniques for successfully resolving difficult decision-making issues. This approach aims to choose the optimal option after taking into account several frequently at-odds factors. In other words, this approach concurrently considers both positive and negative factors (Chakraborty, 2011; Karande and Chakraborty, 2012). The MOORA method has several advantages over some of the decision-making strategies that are already available, including fewer numerical calculations, less computational time, more clarity and more stability when compared to various MCDM techniques like AHP, TOPSIS, ELECTRE and VIKOR (Akkaya et al., 2015; Raj et al., 2023a, d). The MOORA technique consists of the following few steps:

Step 1 – Find the decision matrix X in which x_{ij} shows the performance index of i_{th} alternative concerning j_{th} attribute, $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$ using equation (9).

$$X = [x_{ij}]_{m \times n} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \quad (9)$$

Step 2 – Equation (10) will yield the normalized choice matrix x_{ij}^* .

$$x_{ij}^* = \frac{x_{ij}}{\sqrt{\sum_{j=1}^m x_{ij}^2}}, \forall i, j \quad (10)$$

Step 3 – Each alternative y_i^* overall performance score is determined by combining all favorable criteria and deducting all unfavorable criteria according to equation (11).

$$y_i^* = \sum_{j=1}^q x_{ij}^* - \sum_{j=q+1}^n x_{ij}^* \quad (11)$$

Here, the numbers of advantageous and disadvantageous criteria are q and $(n-q)$, respectively. The various criteria can be given weights using equation (11).

$$y_i^* = \sum_{j=1}^q w_j x_{ij}^* - \sum_{j=q+1}^n w_j x_{ij}^* \quad (12)$$

Step 4 – The y_i^* values from equation (11) and equation (12) are used to rank the possibilities. The MOORA method's ration system computation is shown in the four phases above.

4. Data analysis and results

4.1 Data collection

The construction of the micro-credentials measures and criteria, as indicated in [Table 1](#), followed the effective identification of micro-credentials barriers and their underlying causes from the literature. To gather expert comments from educational institutions and organizations for the evaluation of variables and their comprising criteria, we produced a question survey on the indicators and criteria. The questionnaire was then evaluated with the aid of subject-matter experts. 20 experts from various disciplines were initially contacted. Because Google forms is the most efficient method of data collection and is frequently used by researchers, it was chosen to conduct the survey. The sizes and market shares of these institutions and organizations vary. Only 15 subject-matter experts expressed interest and consented to participate in this study after several reminders. The questionnaire is laid out in the [Appendix](#).

The study's responder size was within the range of MCDM approaches, and the experts were sufficiently competent to have a thorough understanding of the indicators ([Raj et al., 2023c](#)). All 15 specialists are involved in micro-credentialing initiatives and performance evaluation. To create a pair-wise comparison matrix, decision makers (DM) have been asked to convey their opinions using linguistic variables. The statistics are shown in [Table 2](#).

4.2 Evaluating hurdles of micro-credentialing variables

The PROMETHEE approach is being used by [Wang et al. \(2008\)](#) and [Brans et al. \(1986\)](#) to choose the optimal industrial parameters or alternatives. The study considers essential aspects in explaining micro-credentialing factors as well as variables for institutions and organizations and their related criteria. Program Outline (PO), Program Acceptance (PA), Program End Results (PE) and Program Regulation (PR) are the variables used when evaluating the related indicators. [Table 3](#) provides the codes and scale from 1 to 5 for measuring them. Based on these results, the normalized decision matrix in [Table 4](#) (taking step 1 and [equation \(1\)](#) into consideration) and the assessed normalized weighted matrix and evaluated preference functions for each of the variable pair pairs in [Table 5](#) and [Table 6](#) (taking step 3 and [equations \(3\) and \(4\)](#) into consideration) were developed. The aggregate

Profile	Classification	Count
Gender	Female	6
	Male	9
Age	31–40	12
	41–50	2
	Above 50	1
Designation	Mentor	3
	Professional Trainer	4
	Team Leader	2
	Academician (Professors or Teachers)	6
Education	Diploma	0
	Bachelors	2
	Post Graduate	8
	Doctoral in technical education	5
Work experience (in years)	1–7	2
	8–14	5
	15–21	6
	above 21 years	2

Source(s): Authors' own

Table 2.
Respondent's
demographic details

preference function is calculated in the following step and is shown in [Table 7](#) (taking step 4 and [Equation \(5\)](#) into account). Positive and negative flows for data quality characteristics have been measured using step 5, [equations \(6\)](#) and [equation \(7\)](#) (see [Table 8](#)). In [Table 9](#), the net outranking flow rates for these dimensions are displayed (taking step 6 and [Equation \(8\)](#) into account). According to the $\phi(i)$ values presented in [Figure 2](#) for each variable, the charts have been drawn in order of priority. The graphs indicate the final ranking of hurdles for micro-credentialing. The total ranking demonstrates the PROMETHEE method's ranking order selection for the hurdles to micro-variable credentialing's dimensions. Additionally, the industry will be able to increase its effectiveness and maintain its competitiveness in institutions and organizations by using this prioritizing insight.

4.3 Evaluating hurdles of micro-credentialing criteria

[Brauers and Zavadskas \(2006\)](#) proposed the MOORA method as one of the multi-objective optimization techniques for successfully resolving difficult decision-making issues. The

Table 3.
Scale to alternatives
over evaluating
variables and criteria

Linguistic scale	Code	Scale
Unimportant	UI	1
Slightly important	SI	2
Fairly important	FI	3
Important	I	4
Very important	VI	5
Source(s): Authors' own		

Table 4.
Normalized decision
matrix for variables

Variables	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
PO	1	0	1	0.5	1	0	0	0	0	0	0	0.5	0.667	1	0
PA	0	0	1	0	1	0	0	1	0	0	0	0.5	0.667	1	0
PE	1	1	0	1	1	1	1	1	1	1	0.5	0	0	1	1
PR	0	0	1	0.5	0	0	0	0	1	0	1	1	1	0	0
Source(s): Authors' own															

Table 5.
Normalized weighted
matrix for all the pairs
of variables

Variables	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
PO,PA	1	0	0	0.5	0	0	0	-1	0	0	0	0	0	0	0
PO,PE	0	-1	1	-0.5	0	-1	-1	-1	-1	-1	-0.5	0.5	0.667	0	-1
PO,PR	1	0	0	0	1	0	0	0	-1	0	-1	-0.5	-0.33	1	0
PA,PO	-1	0	0	-0.5	0	0	0	1	0	0	0	0	0	0	0
PA,PE	-1	-1	1	-1	0	-1	-1	0	-1	-1	-0.5	0.5	0.667	0	-1
PA,PR	0	0	0	-0.5	1	0	0	1	-1	0	-1	-0.5	-0.33	1	0
PE,PO	0	1	-1	0.5	0	1	1	1	1	1	0.5	-0.5	-0.67	0	1
PE,PA	1	1	-1	1	0	1	1	0	1	1	0.5	-0.5	-0.67	0	1
PE,PR	1	1	-1	0.5	1	1	1	1	0	1	-0.5	-1	-1	1	1
PR,PO	-1	0	0	0	-1	0	0	0	1	0	1	0.5	0.333	-1	0
PR,PA	0	0	0	0.5	-1	0	0	-1	1	0	1	0.5	0.333	-1	0
PR,PE	-1	-1	1	-0.5	-1	-1	-1	-1	0	-1	0.5	1	1	-1	-1
Source(s): Authors' own															

Achieving professional qualifications

Variables	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
PO,PA	0.01	0	0	0.055	0	0	0	0	0	0	0	0	0	0	0
PO,PE	0	0	0.05	0	0	0	0	0	0	0	0	0.025	0.033	0	0
PO,PR	0.01	0	0	0	0.05	0	0	0	0	0	0	0	0	0.05	0
PA,PO	0	0	0	0	0	0	0	0.05	0	0	0	0	0	0	0
PA,PE	0	0	0.05	0	0	0	0	0	0	0	0	0.025	0.033	0	0
PA,PR	0	0	0	0	0.05	0	0	0.05	0	0	0	0	0	0.05	0
PE,PO	0	0.12	0	0.055	0	0.04	0.05	0.05	0.05	0.1	0.06	0	0	0	0.1
PE,PA	0.01	0.12	0	0.11	0	0.04	0.05	0	0.05	0.1	0.06	0	0	0	0.1
PE,PR	0.01	0.12	0	0.055	0.05	0.04	0.05	0.05	0	0.1	0	0	0	0.05	0.1
PR,PO	0	0	0	0	0	0	0	0	0.05	0	0.12	0.025	0.017	0	0
PR,PA	0	0	0	0.055	0	0	0	0	0.05	0	0.12	0.025	0.017	0	0
PR,PE	0	0	0.05	0	0	0	0	0	0	0	0.06	0.05	0.05	0	0

Source(s): Authors' own

Table 6.
Preference functions for all the pairs of variables

Variables	PO	PA	PE	PR
PO	0	0.065	0.108	0.11
PA	0.05	0	0.108	0.15
PE	0.63	0.64	0	0.63
PR	0.217	0.27	0.21	0

Source(s): Authors' own

Table 7.
Aggregate preference function

Variables	Positive flow	Negative flow
PO	0.094333333	0.299
PA	0.102666667	0.325
PE	0.633333333	0.142
PR	0.232333333	0.296666667

Source(s): Authors' own

Table 8.
Positive and negative flows for micro-credentialing variable dimensions

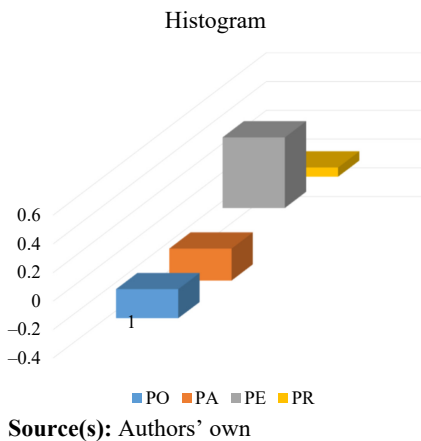
Variables	Net flow	Rank
PO	-0.20467	3
PA	-0.22233	4
PE	0.491333	1
PR	-0.06433	2

Source(s): Authors' own

Table 9.
Net outranking flow values for different dimensions

study considers essential criteria in explaining micro-credentialing factors as well as variables for institutions and organizations. Absence of a strong, unified stakeholder push for micro-credentials (PO1), Absence of a clear definition that can be easily understood without any specific expertise or discipline background (PO2), Absence of a recognized volume of learning (PO3), Pre-designed learning curriculum that are difficult to divide into smaller sections (PO4), Lack of acceptance by one professional or occupational group across multiple

Figure 2.
Rank of the identified
hurdles for micro-
credentialing



jurisdictions (PA1), Lack of acceptance by employers (PA2), Lack of acceptance by individuals (PA3), Mismatch between learning hours and results (PE1), Lack of detailed identity verification measures (PE2), Lack of detailed process assessment methods (PE3), Course descriptions that differ, particularly when every comparable courses are compared, such as learning outcomes (PE4), Unclear remuneration policies (PR1), Lack of trust between different institutions (PR2) and Problem of accreditation and quality assurance is the problem of security issue of credentials (PR 3) are the criteria used when evaluating the related indicators. The normalized decision matrix results obtained by using [equation \(9\)](#) and [equation \(10\)](#) through the MOORA method, which was described in [Section 3](#), are given in [Table 10](#). The overall performance score of each criterion is calculated by adding all beneficial criteria and subtracting the non-beneficial criteria using [equation \(11\)](#) is shown in [Table 11](#). In [Table 12](#), the net outranking flow rates for these criteria are displayed by evaluating y_i^* values (taking step 4 and [Equation \(12\)](#) into account). The total ranking demonstrates the MOORA method's ranking order selection for the criteria of micro-variable credentialing dimensions. According to the y_i^* values presented in [Figure 3](#) for each criterion, the charts have been drawn in order of priority. The graphs indicate the final ranking of associated criteria for micro-credentialing hurdles. Given the data on the criteria describing the various hurdles and challenges affecting the micro-credential program's effectiveness. The data obtained can also be used for developing and implementing measures aimed at prioritizing insight in workplaces.

5. Discussions and findings

This study examined various essential criteria that play a role in explaining micro-credentialing factors and variables for institutions and organizations. These criteria are used to identify and analyze the factors and variables that are hindering the adoption and implementation of micro-credentials in institutions and organizations. By understanding these issues, solutions and recommendations can be developed to overcome the challenges and promote the successful implementations of micro-credentials.

The criteria ranked first belongs to the program outline (PO) and highlights the issue of pre-designed curricula that are difficult to divide into smaller sections (PO4). Micro-credentials are designed to provide more focused learning experiences than traditional degrees or certifications. However, if the course curriculum is not divided systematically into

	DM1	DM2	DM3	DM4	DM5	DM6	DM7	DM8	DM9	DM10	DM11	DM12	DM13	DM14	DM15
PO1	0.31311	0.35218	0.28427	0.2	0.28284	0.33634	0.13736	0.2382	0.30237	0.27343	0.36515	0.43685	0.24434	0.14907	0.17865
PO2	0.25049	0.35218	0.28427	0.26667	0.14142	0.2018	0.13736	0.29775	0.22678	0.27343	0.45644	0.17474	0.12217	0.14907	0.2382
PO3	0.31311	0.26414	0.28427	0.26667	0.14142	0.13453	0.27472	0.17865	0.30237	0.20508	0.27386	0.34948	0.24434	0.14907	0.29775
PO4	0.31311	0.17609	0.35533	0.33333	0.35355	0.26907	0.41208	0.29775	0.22678	0.27343	0.27386	0.26211	0.30542	0.29814	0.29775
PA1	0.31311	0.26414	0.28427	0.33333	0.35355	0.33634	0.13736	0.29775	0.22678	0.20508	0.18257	0.34948	0.30542	0.22361	0.29775
PA2	0.25049	0.26414	0.2132	0.26667	0.21213	0.13453	0.27472	0.29775	0.30237	0.34179	0.09129	0.17474	0.30542	0.29814	0.2382
PA3	0.25049	0.17609	0.2132	0.33333	0.14142	0.2018	0.41208	0.29775	0.30237	0.27343	0.09129	0.17474	0.18325	0.37268	0.29775
PE1	0.18787	0.17609	0.28427	0.33333	0.21213	0.26907	0.27472	0.2382	0.22678	0.34179	0.18257	0.26211	0.30542	0.29814	0.29775
PE2	0.25049	0.17609	0.2132	0.2	0.28284	0.2018	0.41208	0.29775	0.22678	0.27343	0.18257	0.26211	0.30542	0.14907	0.17865
PE3	0.25049	0.17609	0.28427	0.2	0.14142	0.13453	0.13736	0.29775	0.22678	0.20508	0.27386	0.26211	0.30542	0.29814	0.29775
PE4	0.31311	0.26414	0.2132	0.2	0.35355	0.33634	0.13736	0.2382	0.22678	0.27343	0.18257	0.17474	0.24434	0.29814	0.29775
PR1	0.18787	0.26414	0.28427	0.2	0.35355	0.33634	0.13736	0.17865	0.30237	0.27343	0.36515	0.17474	0.24434	0.22361	0.29775
PR2	0.25049	0.35218	0.28427	0.26667	0.21213	0.33634	0.27472	0.2382	0.30237	0.27343	0.27386	0.26211	0.30542	0.29814	0.2382
PR3	0.25049	0.35218	0.2132	0.26667	0.35355	0.33634	0.27472	0.29775	0.30237	0.20508	0.27386	0.26211	0.24434	0.37268	0.2382
Source(s): Authors' own															

Table 10.
Normalized decision
matrix for criteria

Table 11.
Overall performance
score for criteria

	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15
PO1	0.003131	0.042262	0.014213	0.022	0.014142	0.013453	0.006868	0.01191	0.015119	0.027343	0.043818	0.021843	0.012217	0.007454	0.017865
PO2	0.002505	0.042262	0.014213	0.0293	0.007071	0.008072	0.006868	0.014887	0.011339	0.027343	0.054772	0.008737	0.006108	0.007454	0.02382
PO3	0.003131	0.031696	0.014213	0.0293	0.007071	0.005381	0.013736	0.008932	0.015119	0.020508	0.032863	0.017474	0.012217	0.007454	0.029775
PO4	0.003131	0.021131	0.017767	0.0366	0.017678	0.010763	0.020604	0.014887	0.011339	0.027343	0.032863	0.013106	0.015271	0.014907	0.029775
PA1	0.003131	0.031696	0.014213	0.0366	0.017678	0.013453	0.006868	0.014887	0.011339	0.020508	0.021909	0.017474	0.015271	0.01118	0.029775
PA2	0.002505	0.031696	0.01066	0.0293	0.010607	0.005381	0.013736	0.014887	0.015119	0.034179	0.010954	0.008737	0.015271	0.014907	0.02382
PA3	0.002505	0.021131	0.01066	0.0366	0.007071	0.008072	0.020604	0.014887	0.015119	0.027343	0.010954	0.008737	0.009163	0.018634	0.029775
PE1	0.001879	0.021131	0.014213	0.0366	0.010607	0.010763	0.013736	0.01191	0.011339	0.034179	0.021909	0.013106	0.015271	0.014907	0.029775
PE2	0.002505	0.021131	0.01066	0.022	0.014142	0.008072	0.020604	0.014887	0.011339	0.027343	0.021909	0.013106	0.015271	0.007454	0.017865
PE3	0.002505	0.021131	0.014213	0.022	0.007071	0.005381	0.006868	0.014887	0.011339	0.020508	0.032863	0.013106	0.015271	0.014907	0.029775
PE4	0.003131	0.031696	0.01066	0.022	0.017678	0.013453	0.006868	0.01191	0.011339	0.027343	0.021909	0.008737	0.012217	0.014907	0.029775
PR1	0.001879	0.031696	0.014213	0.022	0.017678	0.013453	0.006868	0.008932	0.015119	0.027343	0.043818	0.008737	0.012217	0.01118	0.029775
PR2	0.002505	0.042262	0.014213	0.0293	0.010607	0.013453	0.013736	0.01191	0.015119	0.027343	0.032863	0.013106	0.015271	0.014907	0.02382
PR3	0.002505	0.042262	0.01066	0.0293	0.017678	0.013453	0.013736	0.014887	0.015119	0.020508	0.032863	0.013106	0.012217	0.018634	0.02382
Source(s): Authors' own															

counterfeit micro-credentials are on the rise. This is particularly true for developing nations that have just stepped on to the journey of modern education. This can be corrected and controlled only when the institutions offering micro-credentials are duly accredited and quality-checked. This finding is consistent with the arguments highlighted in [McGreal et al. \(2022\)](#).

The third-ranked criteria also belong to the program regulation (PR) class and highlight the Lack of trust between different institutions (PR2) as one of the significant factors to be considered when delving into a micro-credentialing ecosystem. Micro-credentials may be awarded by a wide range of institutions, from traditional universities to specialized training providers. Without trust between these institutions, it can be difficult to ensure that micro-credentials are being awarded consistently and fairly across different organizations. Since micro-credentialing is a very coarse and organization-specific phenomenon it is a challenge to establish common standards to formulate, present and calibrate course structure across the providing institutions. However, this should not cost learners the time and resources they are putting in to attain the credentials and should be able to use the credentials earned wherever required. This calls for universal validation of the micro-credentials across institutions. Trust is paramount among the participating organizations to lend credibility to the micro-credentialing. This finding builds on the observation made in [Flintberg \(2022\)](#).

The criteria in the fourth position belong to the program outline (PO) class. The absence of a strong, unified stakeholder push for micro-credentials (PO1) is one of the significant factors to be considered while discussing micro-credentialing. This refers primarily to the lack of support and buy-in from key stakeholders, such as educators, employers and students, for the implementation of micro-credentials. Without this support, it can be difficult to develop and implement effective micro-credentialing programs. All the stakeholders have to discuss matters related to micro-credentials and come to a consensus before finalizing any decision. This will help course makers to categorically structure courses that are useful in enhancing skills and addressing the requirements of all the stakeholders involved with the process. [Selvaratnam and Sankey \(2021\)](#) also brought the importance of discussions between the major stakeholders in micro-credentials organizations to the forefront.

Lack of acceptance by one professional or occupational group across multiple jurisdictions (PA1) from the program acceptance (PA) category occupies the fifth position in the overall ranking. This criterion highlights the fact that different professional or occupational groups may have different views on the value and usefulness of micro-credentials, making it difficult to develop programs that are accepted by all groups. This criterion follows from the previously discussed criteria on the lack of trust between institutions. The participating organizations and institutions must share a common ideology on the proposed value of micro-credentialing in reshaping the education eco-system. Without mutual agreement among organizations, the objective of micro-credentials to provide high-quality skill-based training to individuals will fail dramatically. This can lead to limiting the pool of individuals who can earn and use the credentials. It can also make it difficult for employers to recognize and accept the credentials as valid forms of qualifications for certain roles. Also, it can hinder the development of a robust and widely recognized system for micro-credentials, which is necessary for their widespread adoption and use.

Unclear remuneration policies (PR1) in the PR group are ranked sixth in the overall ranking. Ambiguous remuneration policies for instructors offering micro-credential courses can be a significant barrier to the implementation of micro-credentials because it can make it difficult for instructors to understand how they will be compensated for their work. This can discourage instructors from offering micro-credential courses, which can limit the availability of such courses and make it more difficult for individuals to access them. Additionally, unclear remuneration policies can make it difficult for institutions to attract and retain qualified instructors, which can negatively impact the quality of the courses offered. Overall,

unclear remuneration policies can make it difficult to establish and maintain a sustainable ecosystem for micro-credentials. [Vastano et al. \(2022\)](#), [Tinsley et al. \(2022\)](#), [Thi Ngoc Ha et al. \(2022\)](#) confirmed the compensation of instructors as a strong determinant for the implementation of the micro-credentialing landscape.

The criterion in the seventh position in the overall ranking is the Absence of a clear definition that can be easily understood without any specific expertise or discipline background (PO2) in the group program outline (PO). If the definition of a micro-credential is not clearly understood by individuals, employers and other stakeholders, it can make it difficult for them to recognize and accept the credentials as valid forms of qualifications. Another reason why the absence of a clear definition of micro-credentials can be a barrier to their implementation is that it can make it difficult for individuals to understand the value and benefits of earning a micro-credential. If the definition is not easily understandable, it can make it harder for individuals to determine whether a micro-credential is relevant to their career goals or if it will be recognized and valued by potential employers. Furthermore, the lack of a clear definition can also lead to confusion and inconsistency in the quality, content and assessment methods of micro-credentials across different providers and institutions, which can create skepticism among employers and other stakeholders about the credibility and value of micro-credentials, making it difficult to establish a widely recognized and accepted system of micro-credentials.

The criteria in the eighth position are mismatch between learning hours and results (PE1) from the program end-results (PE) group. This criterion refers to the fact that micro-credentials may not always be directly linked to a specific amount of learning or a defined set of learning outcomes, making it difficult to assess the value of the micro-credential and compare it to the other forms of learning and qualifications. Micro-credentials are often based on specific learning outcomes, and if the learning hours required to achieve those outcomes do not match the hours that are typically required for similar credentials, it can create confusion and mistrust among learners and employers. For example, if a micro-credential is designed to demonstrate competency in a specific skill of knowledge area, but the learning hours required to earn the credential are significantly less than the learning hours required for similar credentials, it may be perceived as less valuable or less rigorous by learners and employers. This can make it difficult for individuals to use the credential to advance their careers or for employers to recognize the credential as a valid form of qualification. Additionally, if micro-credentials are based on different learning hour requirements than traditional credentials, it can create challenges in terms of accreditation and recognition by professional bodies, making it difficult for learners to transfer credits and use micro-credentials as stepping stones to pursue further education or career advancement.

The absence of a recognized volume of learning (PO3) in the group program outline (PO) acquires the ninth position in the overall ranking of barriers to implementing micro-credentials. Micro-credentials are often designed to demonstrate specific skills or knowledge, rather than a certain volume of learning. However, this can create confusion among learners and employers because traditional credentials are typically awarded based on the volume of learning, such as the number of credit hours or coursework completed. This observation is in alignment with the findings of [Halas \(2022\)](#).

Course descriptions that differ, particularly when every comparable course is compared, such as learning outcomes (PE4) in the program end-results (PE) group are the tenth criterion identified as a significant barrier in the adoption of micro-credentials in the education landscape. If two micro-credentials are designed to demonstrate competency in a similar skill or knowledge area but have different course descriptions and learning outcomes, it may be difficult for learners to understand the specific skills and knowledge they will be gaining from each credential and for employers to understand how the credentials are different and what they represent. Also, if the course descriptions and learning outcomes vary widely, it can

make it difficult for accreditation bodies to evaluate them and for them to be recognized as valid forms of education.

We focus our attention on the top ten criteria for the current research. However, the last four ranked criteria should also be given due diligence when the decision to adopt micro-credentials is to be taken. Lack of acceptance by employers and individuals (PA2 and PA3) and Lack of detailed process assessment methods and identity verification measures (PE3 and PE2) are some factors that should be carefully considered in the context of adopting micro-credentials. Micro-credentials are a relatively new idea, and some businesses and people may be unfamiliar with them or may not comprehend their significance. Furthermore, if companies and people do not recognize the value of micro-credentials, they may be unwilling to commit the time and money required to obtain them. As a result, it may be difficult for micro-credential providers to attract learners for the micro-credential system to develop traction. Micro-credentials are designed to show certain abilities or information, and it can be difficult to confirm that those who acquire them have truly attained the desired learning goals without suitable evaluation procedures and identity verification processes. Assessment systems must have a clear, consistent and trustworthy approach to assessing and validating skills. It is critical to have a clear, consistent and reliable procedure for evaluating and confirming the skills and information that micro-credentials are supposed to show when it comes to assessment methodologies. Employers and accreditation agencies may find it difficult to identify and accept the reliability of micro-credentials without this. It is critical to have a strong identity verification mechanism in place to confirm that the individual receiving the micro-credential is who they claims to be. This can involve employing secure techniques to authenticate the learner's identity, such as biometric identification or digital certificates. There is a danger of fraud without rigorous identity verification, which undermines the trustworthiness of micro-credentials.

6. Implications

The discussions made in the aforementioned section pave the way for various practical implications. Some of the key takeaways from this study for educators, learners and employers are listed below in the subsection.

6.1 Theoretical implications

It might be difficult to develop micro-credentials that are focused and readily understood by learners if the course content is not broken into smaller portions. This is especially true for courses that are large and complex since breaking them up into smaller pieces might result in knowledge loss or too much clutter for learners to absorb in a short amount of time. This emphasizes the need for meticulous planning and design when developing micro-credentials to ensure that they properly connect with specified learning goals and are readily understood by learners. Without certification and quality assurance systems in place, it can be difficult to identify whether micro-credentials are issued by credible organizations and whether they accurately represent the knowledge and abilities of those who obtain them. This can lead to a proliferation of counterfeit micro-credentials, especially in poor countries where standards and control may be lax. To overcome this issue, institutions that offer micro-credentials must be properly certified and have quality assurance mechanisms in place to check the correctness, relevance and validity of the micro-credentials they grant. This can serve to raise the legitimacy and value of micro-credentials, as well as assure their acceptance as legitimate forms of knowledge and skill certification. Lack of trust between the participating institutions offering micro-credentials should be overcome by establishing universal standards to gauge the quality and accuracy of the courses being offered under the flagship of micro-

credentialing. This can be made possible only by increasing collaborations and information sharing between the institutions and stakeholders on matters about designing courses, formulating grading schemes and structuring compensation policies for instructors.

Unclear remuneration policies for instructors can significantly impact the quality of courses offered in micro-credential platforms. Inadequate remunerations can discourage quality instructors from offering courses which is detrimental for learners with intended learning outcomes. Institutions can find it extremely difficult to find quality educators in the absence of good remuneration policies for qualified instructors. Thus, it is necessary to provide a good salary as per the standards that motivate quality instructors to participate in offering courses in micro-credential platforms. It is pivotal to design the courses in a micro-credential platform including its purpose, content and assessment methods in easy language that can be readily comprehended by educators, learners and employers. Ensuring consistency in micro-credentials across institutions through properly established guidelines can lead to strengthening the credibility of certifications provided by micro-credentials. Extensive research should be conducted with all the stakeholders including educators, learners and employers to clearly understand their requirements and expectations from the micro-credential courses and subsequently, the micro-credentials should be developed. Providing video resources and brochures to help individuals understand the value and benefits of micro-credentials and guide them in attaining their career goals. The learning hours and learning outcomes required for each micro-credential should be aligned with the industry standards and requirements of similar credentials. There should be a dedicated system for the assessment and evaluation of micro-credentials to ensure that the learning hours, learning outcomes and volume of learning are properly matched with the requirements of the stakeholders. Adequate opportunities should be created for individuals to gauge their performance on micro-credentials through formative assessments, progress reports and portfolio reviews. Due consideration should be given to allowing individuals to earn micro-credentials in a flexible manner, such as through self-paced and modular learning, to better suit their needs and the available learning hours. The volume of learning should be properly communicated to potential employers and stakeholders so that the value and rigor of micro-credentials can be properly assessed. To effectively measure the volume of learning and make micro-credentials more comparable, various established metrics such as the European Credit Transfer and Accumulation System (ECTS) or credit hours should be used in the micro-credential landscape. ECTS is a framework used to measure the volume of learning in different educational programs across Europe.

6.2 Practical implications

To create micro-credentials that learners can simply understand, careful planning and design are essential. To accomplish this, course material must be divided into manageable chunks, learning objectives must be met, and knowledge loss or overwhelming clutter must be avoided. It ensures that students effectively understand the topic and that the micro-credentials relate to the goals of their education. Institutions that offer micro-credentials must set up quality control processes and receive the appropriate certification. This guarantees that the micro-credentials granted by these organizations appropriately reflect the learners' knowledge and abilities. Micro-credentials' legitimacy and worth are increased by certification and quality control, which helps to advance their recognition as acceptable types of knowledge and skill certification. For uniform standards in micro-credentialing to be established, cooperation and information sharing across institutions and stakeholders are essential. Collectively defining standards and criteria for course quality, grading systems and remuneration policies is something institutions may do by working together. All parties involved in the micro-credential environment gain from this cooperative effort's promotion of

uniformity, assurance of confidence and ability to create a strong foundation. For micro-credential platforms to successfully recruit and retain top instructors, adequate compensation practices are crucial. Competent educators are encouraged to participate when paid competitively, which improves the caliber of courses and eventually benefits students. Furthermore, by creating criteria for consistency in micro-credentials across institutions, uniformity in standards is ensured, enhancing the legitimacy and comparability of certificates in the micro-credential environment.

7. Conclusions, limitations and future scope of research

This study investigated the primary factors influencing the implementation of micro-credentials and prioritizing those using ranking methods such as PROMETHEE and MOORA. Both methods were employed to identify the rank order selection for the hurdles and criteria of micro-credentialing dimensions. Among the main criteria identified as exerting significant influence in the implementation of micro-credentials were providing clear definitions and guidelines for micro-credentials, structuring of micro-credential courses, unclear remuneration policies, lack of trust among participating institutions, ensuring consistency in the quality, content and assessment methods of micro-credentials across different providers, working to build trust and credibility among employers and other stakeholders, lack of acceptance by employers and individuals, and lack of detailed identity verification measures. The barriers identified in this study can be used by educational institutions, organizations and policymakers to better understand the issues impeding the adoption of micro-credentials and accordingly develop strategies to address them subsequently helping to make micro-credentials a more widely recognized and accepted form of education and qualifications. Furthermore, it is to be noted that micro-credential is still a relatively new concept, and particularly, in developing countries it is yet to find a strong ground to stand. Therefore, it is crucial to conduct regular evaluations of the identified barriers and challenges in the context of the adoption of micro-credentials.

Certain important limitations can be viewed as future research opportunities in the micro-credential landscape. This research is based on a specific study that used PROMETHEE and MOORA methods, and the results may not be generalizable to other studies or contexts. This is an exploratory study involving some key variables in micro-credentialing, however, other variables can also be selected based on experts' opinions. The findings are based on the data and information available at the time of the study and may not account for new developments and changes in the micro-credential landscape. The study lacks the cost-benefit analysis of the proposed solutions to address the barriers, it may be important to consider the cost of implementing the solutions compared to the benefits of addressing the barriers. Additionally, the scalability of the proposed solutions has not been considered in this study. Some solutions might be difficult to implement on a large scale and may not be sensible for certain institutions and organizations.

References

- Ahmat, N.H.C., Bashir, M.A.A., Razali, A.R. and Kasolang, S. (2021), "Micro-credentials in higher education institutions: challenges and opportunities", *Asian Journal of University Education*, Vol. 17 No. 3, pp. 281-290, doi: [10.24191/ajue.v17i3.14505](https://doi.org/10.24191/ajue.v17i3.14505).
- Akkaya, G., Turanoğlu, B. and Öztaş, S. (2015), "An integrated fuzzy AHP and fuzzy MOORA approach to the problem of industrial engineering sector choosing", *Expert Systems with Applications*, Vol. 42 No. 24, pp. 9565-9573, doi: [10.1016/j.eswa.2015.07.061](https://doi.org/10.1016/j.eswa.2015.07.061).

- AlHamad, N.S. and AlAmri, K. (2021), "The association between social media use and depressive symptoms among adults in Riyadh, Saudi Arabia", *Journal of Family Medicine and Primary Care*, Vol. 10 No. 9, p. 3336.
- Athawale, V.M. and Chakraborty, S. (2010), "Facility location selection using PROMETHEE II method", *Proceedings of the 2010 International Conference on Industrial Engineering and Operations Management* Dhaka, pp. 59-64.
- Aydarova, E. (2021), "NEPC review: harnessing micro-credentials for teacher growth", National Education Policy Center, Boulder, CO, available at: <http://nepc.colorado.edu/thinktank/microcredentials> (accessed 3 January 2024).
- Aydarova, E. (2022), "Shadow elite of transnational policy networks: intermediary organizations and the production of teacher education policies", in *The Palgrave Handbook of Teacher Education Research*, pp. 1-21.
- Barry, W. (2022), "The role of 'knowledgeable others' in supporting academics' professional learning: implications for academic development", *Perspectives: Policy and Practice in Higher Education*, Vol. 27, pp. 1-10, doi: [10.1080/13603108.2022.2131650](https://doi.org/10.1080/13603108.2022.2131650).
- Bigelow, A., Booth, C., Brockerhoff-Macdonald, B., Cormier, D., Dinsmore, C., Grey, S., Harrison, L., Hobbs, A., Lee, S., Maher, P., McArthur, F., Mitchell-Ashley, T., Mosley, J., Papple, J., Porter, J., Presant, D., Sommer, J. and Zahedi, E. (2022), "eCampusOntario's micro-credential toolkit", available at: <https://openlibrary.ecampusontario.ca/item-details/#/3b7a4bed-0fa3-4b9d-bd02-4bcfdb919090>
- Boud, D. and Jorre de St Jorre, T. (2021), "The move to micro-credentials exposes the deficiencies of existing credentials", *Journal of Teaching and Learning for Graduate Employability*, Vol. 12 No. 1, pp. 18-20, doi: [10.21153/jtlge2021vol12no1art1023](https://doi.org/10.21153/jtlge2021vol12no1art1023).
- Brans, J.P., Vincke, P. and Mareschal, B. (1986), "How to select and how to rank projects: the PROMETHEE method", *European Journal of Operational Research*, Vol. 24 No. 2, pp. 228-238, doi: [10.1016/0377-2217\(86\)90044-5](https://doi.org/10.1016/0377-2217(86)90044-5).
- Brauers, W.K. and Zavadskas, E.K. (2006), "The MOORA method and its application to privatization in a transition economy", *Control and Cybernetics*, Vol. 35 No. 2, pp. 445-469.
- Brown, M., Mhichil, M.N.G., Beirne, E. and Mac Lochlainn, C. (2021), "The global micro-credential landscape: charting a new credential ecology for lifelong learning", *Journal of Learning for Development*, Vol. 8 No. 2, pp. 228-254, doi: [10.56059/jl4d.v8i2.525](https://doi.org/10.56059/jl4d.v8i2.525).
- Brunello, G. and Wruuck, P. (2021), "Skill shortages and skill mismatch: a review of the literature", *Journal of Economic Surveys*, Vol. 35 No. 4, pp. 1145-1167, doi: [10.1111/joes.12424](https://doi.org/10.1111/joes.12424).
- Carmichael, J.J. and Eaton, S.E. (2023), "Admissions fraud in Canadian higher education", in *Fake Degrees and Fraudulent Credentials in Higher Education*, Springer International Publishing, Cham, pp. 23-48.
- Chakraborty, S. (2011), "Applications of the MOORA method for decision making in manufacturing environment", *The International Journal of Advanced Manufacturing Technology*, Vol. 54 No. 9, pp. 1155-1166, doi: [10.1007/s00170-010-2972-0](https://doi.org/10.1007/s00170-010-2972-0).
- Charania, A. (2022), "Workshops to digital badges: a dynamic journey of teacher professional development in ITE", in *Integrated Approach to Technology in Education in India*, Routledge India, pp. 84-130.
- Clausen, J.M. (2022), "Learning to fly: development and design of a micro-credentialing system for an educator preparation program in the absence of a required educational technology course", *TechTrends*, Vol. 66 No. 2, pp. 276-286, doi: [10.1007/s11528-021-00673-x](https://doi.org/10.1007/s11528-021-00673-x).
- Cook, E. (2021), "Practice-based engineering: mathematical competencies and micro-credentials", *International Journal of Research in Undergraduate Mathematics Education*, Vol. 7 No. 2, pp. 284-305, doi: [10.1007/s40753-020-00128-3](https://doi.org/10.1007/s40753-020-00128-3).
- Curtis, H., Hogeveen, B., Kang, J., Le Thu, H., Rajagopalan, R.P. and Ray, T. (2022), "Digital Southeast Asia: opportunities for Australia-India cooperation to support the region in the post-COVID-19

- context", available at: <https://policycommons.net/artifacts/2242963/digital-southeast-asia/3001066/> (accessed 3 January 2024).
- Dawson, N.J. (2021), "Changing labour market dynamics in Australia: skill shortages, job transitions, and artificial intelligence technology adoption", (Doctoral dissertation).
- Del Toro, H. (2021), "Liberal arts micro-credential: teaching a police officer reflection, empathy, and self-awareness; going beyond training a police officer to act and react", (Doctoral dissertation, DePaul University).
- De Monte, J. (2017), "Micro-credentials for teachers: what three early adopter states have learned so far", American Institutes for Research, available at: <https://www.air.org/sites/default/files/downloads/report/Micro-Credentials-for-Teachers-September-2017.pdf>
- Doumpos, M. and Zopounidis, C. (2004), "A multicriteria classification approach based on pairwise comparisons", *European Journal of Operational Research*, Vol. 158 No. 2, pp. 378-389, doi: [10.1016/j.ejor.2003.06.011](https://doi.org/10.1016/j.ejor.2003.06.011).
- Duklas, J. (2020), "Micro-credentials: trends in credit transfer and credentialing", *British Columbia Council on Admissions and Transfer*, European Education Area, available at: <https://education.ec.europa.eu/education-levels/higher-education/micro-credentials> (accessed 5 June 2023).
- European Education Area (2023), available at: <https://education.ec.europa.eu/education-levels/higher-education/micro-credentials> (accessed 3 January 2024).
- Fedock, B., Kebritchi, M., Sanders, R. and Holland, A. (2016), "Digital badges and micro-credentials: digital age classroom practices, design strategies, and issues", *Foundation of Digital Badges and Micro-Credentials: Demonstrating and Recognizing Knowledge and Competencies*, Springer, Cham, pp. 273-286, doi: [10.1007/978-3-319-15425-1_15](https://doi.org/10.1007/978-3-319-15425-1_15).
- Finkelstein, J., Knight, E. and Manning, S. (2013), *The Potential and Value of Using Digital Badges for Adult Learners*, American Institutes for Research, Washington, DC, available at: https://lincs.ed.gov/publications/pdf/AIR_Digital_Badge_Report_508.pdf
- Flintberg, B. (2022), "Digital micro-credentials for upskilling and reskilling in the vocational sector: a study on the potential use of a blockchain-based ICT system for micro-credentials in Sweden", available at: <https://lnu.diva-portal.org/smash/record.jsf?pid=diva2%3A1681162&dsid=5398> (accessed 3 January 2024).
- Fogarty, S., Cunningham, C., Striepe, M. and Rhodes, D. (2023), "The provision of vocational education and training within contemporary political socio-economic conditions: a review of the literature", *International Journal of Training Research*, Vol. ahead-of-print No. ahead-of-print, pp. 1-18, doi: [10.1080/14480220.2023.2263671](https://doi.org/10.1080/14480220.2023.2263671).
- Forbes (2023), available at: <https://www.forbes.com/sites/niallmccarthy/2016/04/18/the-countries-facing-the-greatest-skill-shortages-infographic/?sh=6aceda721b56> (accessed 5 June 2023).
- Frederiksen, L. (2013), "Digital badges", *Public Services Quarterly*, Vol. 9 No. 4, pp. 321-325, doi: [10.1080/15228959.2013.842414](https://doi.org/10.1080/15228959.2013.842414).
- Garg, P., Raj, R., Kumar, V., Singh, S., Pahuja, S. and Sehrawat, N. (2023), "Elucidating the role of consumer decision making style on consumers' purchase intention: the mediating role of emotional advertising using PLS-SEM", *Journal of Economy and Technology*, Vol. 1, pp. 108-118, doi: [10.1016/j.ject.2023.10.001](https://doi.org/10.1016/j.ject.2023.10.001).
- Ghasia, M., Machumu, H. and Smet, E. (2019), "Micro-credentials in higher education institutions: an exploratory study of its place in Tanzania", *International Journal of Education and Development Using ICT*, Vol. 15 No. 1, pp. 219-230.
- Gigauri, I. (2021), "New economic concepts shaping business models in post-pandemic era", *International Journal of Innovative Technologies in Economy*, Vol. 1 No. 33, pp. 1-12, doi: [10.31435/rsglobal_ijite/30032021/7393](https://doi.org/10.31435/rsglobal_ijite/30032021/7393).
- Gordon, S., Yough, M., Finney-Miller, E., Mathew, S., Haken-Hughes, A. and Ariati, J. (2021), "Faculty perceptions of teaching diversity: definitions, benefits, draw-backs, and barriers", *Current Psychology*, Vol. 40 No. 2, pp. 1-10, available at: <https://www.researchgate.net/publication/>

- [373821228_Desafios_de_la_interculturalidad_en_la_formacion_de_profesores_de_Educacion_Fisica_Challenges_of_interculturality_in_pre-service_physical_education_teachers_training#fullTextFileContent](#) (accessed 3 January 2024).
- Hajkowicz, S. and Higgins, A. (2008), "A comparison of multiple criteria analysis techniques for water resource management", *European Journal of Operational Research*, Vol. 184 No. 1, pp. 255-265, doi: [10.1016/j.ejor.2006.10.045](#).
- Halas, A.S. (2022), "Mapping european eco-system of micro-credentials in higher education", (Doctoral dissertation, Sumy State University).
- Hu, D., Yuan, B., Luo, J. and Wang, M. (2021), "A review of empirical research on ICT applications in teacher professional development and teaching practice", *Knowledge Management and E-Learning: An International Journal*, Vol. 13 No. 1, pp. 1-20.
- Hunt, T., Carter, R., Zhang, L. and Yang, S. (2020), "Micro-credentials: the potential of personalized professional development", *Development and Learning in Organizations: An International Journal*, Vol. 34 No. 2, pp. 33-35, doi: [10.1108/dlo-09-2019-0215](#).
- Huynh-Cam, T.T., Kumar, V., Nguyen, V.C., Le-Thi, B.V. and Huynh, L.B.K. (2022), "The evaluation of students' satisfaction with instructor performance having major key factors: the case of a Vietnamese university", *International Journal of Knowledge and Learning*, Vol. ahead-of-print No. ahead-of-print, pp. 1-24, doi: [10.1504/IJKL.2024.135410](#).
- Ifenthaler, D., Bellin-Mularski, N. and Mah, D.K. (2016), *Foundation of Digital Badges and Micro-credentials*, Springer International Publishing.
- Jääskelä, P., Heilala, V., Kärkkäinen, T. and Häkkinen, P. (2021), "Student agency analytics: learning analytics as a tool for analysing student agency in higher education", *Behaviour and Information Technology*, Vol. 40 No. 8, pp. 790-808, doi: [10.1080/0144929x.2020.1725130](#).
- Jamaluddin, J.E., Hamid, F.A. and Aziz, T.I.A. (2021), "Micro-credential: introduction to communication network", *Innovating Education for a Better Tomorrow, IUCEL2022 Proceedings*, pp. 173-176.
- Karande, P. and Chakraborty, S. (2012), "Application of multi-objective optimization on the basis of ratio analysis (MOORA) method for materials selection", *Materials and Design*, Vol. 37, pp. 317-324, doi: [10.1016/j.matdes.2012.01.013](#).
- Kato, S., Galán-Muros, V. and Weko, T. (2020), "The emergence of alternative credentials", OECD Education Working Papers No. 216, OECD Publishing, Paris, available at: https://www.oecd-ilibrary.org/education/the-emergence-of-alternative-credentials_b741f39e-en
- Kohler, M., Gamrat, C., Raish, V. and Gross, E. (2021), "Microlearning and micro-credentials in higher education", in *Microlearning in the Digital Age*, Routledge, pp. 109-128.
- Kumar, J.A., Richard, R.J., Osman, S. and Lowrence, K. (2022a), "Micro-credentials in leveraging emergency remote teaching: the relationship between novice users' insights and identity in Malaysia", *International Journal of Educational Technology in Higher Education*, Vol. 19 No. 1, pp. 1-23, doi: [10.1186/s41239-022-00323-z](#).
- Kumar, V., Mittal, A., Sharma, A. and Ahmad, R. (2022b), "The role of stress personalities: a comparative study between Indian academia and industry", *Journal of Mental Health, Training, Education and Practice*, Vol. 17 No. 6, pp. 506-525, doi: [10.1108/jmhtep-07-2021-0081](#).
- LaMagna, M. (2017), "Placing digital badges and micro-credentials in context", *Journal of Electronic Resources Librarianship*, Vol. 29 No. 4, pp. 206-210, doi: [10.1080/1941126x.2017.1378538](#).
- Lexman, R.R., John, J. and Friedler, A. (2020), "Campus-IL: enhancing teachers' learning experience", *South Asian Journal of Management*, Vol. 27 No. 2, pp. 189-216.
- Lim, C.L., Nair, P.K., Keppell, M.J., Hassan, N. and Ayub, E. (2018), "December Developing a framework for the university-wide implementation of micro-credentials and digital badges: a case study from a Malaysian private university", *2018 IEEE 4th International Conference on Computer and Communications (ICCC)*, pp. 1715-1719, IEEE.
- MacKinnon, T. (2021), "Open badges", in *Innovative Language Pedagogy Report*, pp. 57-62.

- Mardisentosa, B., Rahardja, U., Zelina, K., Oganda, F.P. and Hardini, M. (2021), "Sustainable learning micro-credential using blockchain for student achievement records", *2021 Sixth International Conference on Informatics and Computing (ICIC)*, pp. 1-6, IEEE.
- Martinez-Marroquin, E. and Male, S. (2021), "Micro-credentials for recognition of workplace learning: provocation", *Journal of Teaching and Learning for Graduate Employability*, Vol. 12 No. 1, pp. 52-57.
- Mashford-Pringle, A., Tan, S., Stutz, S. and Tjong, G. (2023), "Designing accountability measures for health professionals: results from a community-based micro-credential: case study on Indigenous cultural safety", *BMC Public Health*, Vol. 23 No. 1, pp. 1-10, doi: [10.1186/s12889-023-15721-9](https://doi.org/10.1186/s12889-023-15721-9).
- McCartney, M.L. and Rick, H.A. (2021), "The adult student/consumer model: micro-credentials as a solution for adult learners", in *Ensuring Adult and Non-traditional Learners' Success with Technology, Design, and Structure*, IGI Global, pp. 244-261.
- McGreal, R. and Olcott, D. (2022), "A strategic reset: micro-credentials for higher education leaders", *Smart Learning Environments*, Vol. 9 No. 1, pp. 1-23, doi: [10.1186/s40561-022-00190-1](https://doi.org/10.1186/s40561-022-00190-1).
- McGreal, R., Mackintosh, W., Cox, G. and Olcott, D. (2022), "Bridging the gap: micro-credentials for development: UNESCO chairs policy brief form-under the III world higher education conference (WHEC 2021) type: collective X", *International Review of Research in Open and Distributed Learning*, Vol. 23 No. 3, pp. 288-302, doi: [10.19173/irrodl.v23i3.6696](https://doi.org/10.19173/irrodl.v23i3.6696).
- Messaoud, S.B., Ilahi, M. and Cheniti-Belcadhi, L. (2022), "Towards a personalized micro-credentials approach based on learning analytics for reducing the gap university-industry", *2022 IEEE Global Engineering Education Conference (EDUCON)*, IEEE, pp. 1448-1454.
- Milligan, S. and Kennedy, G. (2017), "To what degree? Alternative micro-credentialing in a digital age", in James, R., French, S. and Kelly, P. (Eds), *Visions for the Future of Australian Tertiary Education*, 1st ed., Centre for the Study of Higher Education, University of Melbourne, pp. 41-54.
- Msweli, N.T., Twinomurinzi, H. and Ismail, M. (2022), "The international case for micro-credentials for life-wide and life-long learning: a systematic literature review", *Interdisciplinary Journal of Information, Knowledge, and Management*, Vol. 17, pp. 151-190, doi: [10.28945/4954](https://doi.org/10.28945/4954).
- Murphy, C. and O'Sullivan, M. (2021), "Running to stand still? Two decades of trade union activity in the Irish long-term care sector", *Transfer: European Review of Labour and Research*, Vol. 27 No. 3, pp. 383-397, doi: [10.1177/10242589211028461](https://doi.org/10.1177/10242589211028461).
- Navanitha, M., Savita, K.S., Arshad, N.I., Isawasan, P., Adams, D., Ahmat, N.H.C. and Shariman, T.P.N.B.T. (2022), "Survey instrument validation for a learner's motivation to attain digital badge for micro-credential courses: cognitive interview and validity test", *Journal of Hunan University Natural Sciences*, Vol. 49 No. 10, pp. 77-87, doi: [10.55463/issn.1674-2974.49.10.9](https://doi.org/10.55463/issn.1674-2974.49.10.9).
- Olcott, D. Jr (2022), "Micro-credentials: a catalyst for strategic reset and change in us higher education", *American Journal of Distance Education*, Vol. 36 No. 1, pp. 19-35, doi: [10.1080/08923647.2021.1997537](https://doi.org/10.1080/08923647.2021.1997537).
- Paliwal, M., Raj, R., Kumar, V., Singh, S., Sharma, N.K., Suri, A. and Kumari, M. (2010), "Informal workers in India as an economic shock absorber in the era of COVID-19: a study on policies and practices", in *Human Systems Management*, pp. 1-20, (Preprint), doi: [10.3233/hsm-220155](https://doi.org/10.3233/hsm-220155).
- Pirkkalainen, H., Sood, I., Padron Napoles, C., Kukkonen, A. and Camilleri, A. (2023), "How might micro-credentials influence institutions and empower learners in higher education?", *Educational Research*, Vol. 65 No. 1, pp. 40-63, doi: [10.1080/00131881.2022.2157302](https://doi.org/10.1080/00131881.2022.2157302).
- Quigley, S.J., Lui, E., Whelpley, S. and Flot, J. (2022), "GitHub in the technical and professional communications classroom", *Reflections*, Vol. 22 No. 1, pp. 242-274.
- Raj, R., Kumar, V. and Verma, P. (2023a), "Big data analytics in mitigating challenges of sustainable manufacturing supply chain", *Operations Management Research*, Vol. 16 No. 4, pp. 1-15, doi: [10.1007/s12063-023-00408-6](https://doi.org/10.1007/s12063-023-00408-6).

- Raj, R., Kumar, V., Sharma, N.K., Singh, S., Mahlawat, S. and Verma, P. (2023b), "The study of remote working outcome and its influence on firm performance", *Social Sciences and Humanities Open*, Vol. 8 No. 1, 100528, doi: [10.1016/j.ssho.2023.100528](https://doi.org/10.1016/j.ssho.2023.100528).
- Raj, R., Singh, A., Kumar, V. and Verma, P. (2023c), "Analyzing the potential benefits and use cases of ChatGPT as a tool for improving the efficiency and effectiveness of business operations", *BenchCouncil Transactions on Benchmarks, Standards and Evaluations*, Vol. 3 No. 3, 100140, doi: [10.1016/j.tbench.2023.100140](https://doi.org/10.1016/j.tbench.2023.100140).
- Raj, R., Kumar, V. and Shah, B. (2023d), "Big data analytics adaptive prospects in sustainable manufacturing supply chain", *Benchmarking: An International Journal*, Vol. ahead-of-print No. ahead-of-print, doi: [10.1108/BIJ-11-2022-0690](https://doi.org/10.1108/BIJ-11-2022-0690).
- Rajak, B.K., Raj, R., Kumar, V., Singh, P., Verma, P., Mahlawat, S., Singh, S. and Reddy, K.V. (2023), "Torn ties and waning morale: unravelling the link between family incivility, employee engagement and perceived supervisor support", *Social Sciences and Humanities Open*, Vol. 8 No. 1, 100732.
- Randall, D.L. and West, R.E. (2022), "Who cares about open badges? An examination of principals' perceptions of the usefulness of teacher open badges in the United States", *Open Learning: The Journal of Open, Distance and E-Learning*, Vol. 37 No. 1, pp. 65-83, doi: [10.1080/02680513.2020.1752166](https://doi.org/10.1080/02680513.2020.1752166).
- Resei, C., Friedl, C. and Žur, A. (2018), "MOOCs and entrepreneurship education-contributions, opportunities and gaps", *International Entrepreneurship Review*, Vol. 4 No. 3, p. 151.
- Rieder, A.T. (2022), "Using micro-credentials and digital badges in education and training: a literature review", *Society for Information Technology & Teacher Education International Conference*, pp. 195-200, Association for the Advancement of Computing in Education (AACE).
- Rosenbaum, M.S., Russell-Bennett, R. and Contreras-Ramírez, G. (2021), "Business education in profound disruption", *Journal of Services Marketing*, Vol. 35 No. 5, pp. 553-558, doi: [10.1108/jsm-08-2021-528](https://doi.org/10.1108/jsm-08-2021-528).
- Rossiter, D. and Tynan, B. (2019), "Designing and implementing micro-credentials: a guide for practitioners", Commonwealth of Learning, available at: <http://oasis.col.org/handle/11599/3279> (accessed 3 January 2024).
- Rottmann, A.K. and Duggan, M.H. (2021), "Micro-credentials in higher education", *Handbook of Research on Innovations in Non-traditional Educational Practices*, IGI Global, pp. 223-236.
- Ruddy, C. and Ponte, F. (2019), "Preparing students for university studies and beyond: a micro-credential trial that delivers academic integrity awareness", *Journal of the Australian Library and Information Association*, Vol. 68 No. 1, pp. 56-67.
- Selvaratnam, R.M. and Sankey, M.D. (2021), "An integrative literature review of the implementation of micro-credentials in higher education: implications for practice in Australasia", *Journal of Teaching and Learning for Graduate Employability*, Vol. 12 No. 1, pp. 1-17, doi: [10.21153/jtlge2021vol12no1art942](https://doi.org/10.21153/jtlge2021vol12no1art942).
- Singh, S., Paliwal, M., Kumar, V., Singh, S.V.P. and Mahlawat, S. (2023), "Explaining the linkage between antecedents' factors of adopting online classes and perceived learning outcome using extended UTAUT model", *Data and Information Management*, Vol. 7 No. 4, 100052, Ahead-of-Print, doi: [10.1016/j.dim.2023.100052](https://doi.org/10.1016/j.dim.2023.100052).
- Tereseviciene, M., Trepule, E., Dauksiene, E., Tamoliune, G. and Costa, N. (2020), "Are universities ready to recognize open online learning?", *International Education Studies*, Vol. 13 No. 2, pp. 21-32, doi: [10.5539/ies.v13n2p21](https://doi.org/10.5539/ies.v13n2p21).
- Thi Ngoc Ha, N., Spittle, M., Watt, A. and Van Dyke, N. (2022), "A systematic literature review of micro-credentials in higher education: a non-zero-sum game", *Higher Education Research and Development*, pp. 1-22.
- Tinsley, B., Cacicio, S., Shah, Z., Parker, D., Onenge, O. and Luke Luna, C. (2022), "Micro-credentials for social mobility in rural postsecondary communities: a landscape report", Digital Promise.

- Varadarajan, S., Koh, J.H.L. and Daniel, B.K. (2023), "A systematic review of the opportunities and challenges of micro-credentials for multiple stakeholders: learners, employers, higher education institutions and government", *International Journal of Educational Technology in Higher Education*, Vol. 20 No. 1, pp. 1-24, doi: [10.1186/s41239-023-00381-x](https://doi.org/10.1186/s41239-023-00381-x).
- Vastano, S.E., Scott-Covington, R., Dusinberre, J., Matejka, A. and Feistman, R. (2022), "Micro-credentialing teacher leadership: coaching digital professional development", in *Handbook of Research on Credential Innovations for Inclusive Pathways to Professions*, IGI Global, pp. 356-374.
- Verma, P., Kumar, V., Mittal, A., Rathore, B., Jha, A. and Rahman, M. (2023), "The role of 3S in big data quality: a perspective on operational performance indicators using an integrated approach", *The TQM Journal*, Vol. 35 No. 1, pp. 153-182, doi: [10.1108/tqm-02-2021-0062](https://doi.org/10.1108/tqm-02-2021-0062).
- Wang, T.C., Chen, L.Y. and Chen, Y.H. (2008), "Applying fuzzy PROMETHEE method for evaluating IS outsourcing suppliers", *2008 Fifth International Conference on Fuzzy Systems and Knowledge Discovery*, Vol. 3, pp. 361-365, IEEE.
- Ward, R., Phillips, O., Bowers, D., Crick, T., Davenport, J.H., Hanna, P., Hayes, A., Irons, A. and Prickett, T. (2021), "Towards a 21st century personalised learning skills taxonomy", *IEEE Global Engineering Education Conference (EDUCON)*, pp. 344-354, doi: [10.1109/EDUCON46332.2021.9453883](https://doi.org/10.1109/EDUCON46332.2021.9453883).
- West, R.E. and Cheng, Z. (2022), "Digital credential evolution", in *Handbook of Open, Distance and Digital Education*, Springer, available at: <https://edtechbooks.org/-NJwY>
- Wheelahan, L. and Moodie, G. (2021), "Analysing micro-credentials in higher education: a Bernsteinian analysis", *Journal of Curriculum Studies*, Vol. 53 No. 2, pp. 212-228.
- White, S. (2021), "Developing credit based micro-credentials for the teaching profession: an Australian descriptive case study", *Teachers and Teaching*, Vol. 27 No. 7, pp. 696-711, doi: [10.1080/13540602.2021.2003324](https://doi.org/10.1080/13540602.2021.2003324).
- Wolz, E., Gottlieb, M. and Pongratz, H. (2021), "Digital credentials in higher education institutions: a literature review", *International Conference on Wirtschaftsinformatik*, pp. 125-140, Springer, Cham.
- Yanek, J.J. (2020), "Digital badges: a study of K-12 educators' degree of exposure to 21st century skills developed through micro-credentials", (Doctoral dissertation, Delaware Valley University).
- Zain, S. (2023), "Micro-credentials: need to be benchmarked across institutions", in *Benchmarking Library, Information and Education Services*, Chandos Publishing, pp. 329-338.
- Zou, H., Ullah, A., Qazi, Z., Naeem, A. and Rehan, S. (2023), "Impact of micro-credential learning on students' perceived employability: the mediating role of human capital", *International Journal of Educational Management*, Vol. ahead-of-print No. ahead-of-print, pp. 1-19, doi: [10.1108/ijem-01-2023-0002](https://doi.org/10.1108/ijem-01-2023-0002).

Appendix Questionnaire

This is a questionnaire regarding examining the factors impeding the implementation of micro-credentials and accepting it as a credible source of earning professional qualifications and certifications necessary for pursuing higher education or other career goals. This survey focuses on finding out the major barriers/challenges and sub barriers/challenges to mitigate them and develop a comprehensive structural relationship to rank them to streamline the functions of educational institutions, organizations, and policymakers to better understand the issues and develop strategies to address them, making micro-credentials a more recognized form of education and qualifications using PROMETHEE and MOORA methods. We assure you that the information shared by you for this research will be treated as highly confidential and will be used strictly for this research work only. Kindly spare a few minutes and complete this questionnaire.

SECTION 1: Demographic details

Please take a few moments to answer the following questions that will assist in making inferences regarding the population of this study.

Please take a few moments to answer the following questions that will assist in making inferences regarding the population of this study.

- (1) Gender
 - Male
 - Female
- (2) Age
 - 31–40
 - 41–50
 - Above 50
- (3) Your Designation
 - Mentor
 - Professional trainer
 - Team leader
 - Academicians (professors or teachers)
- (4) Highest level of Education
 - Diploma
 - Bachelors
 - Post-graduate
 - Doctoral in technical education
- (5) Current Organizational Tenure
 - 1–7 years
 - 8–14 years
 - 15–21 years
 - Above 21 years

SECTION 2: Major barriers

SN	Challenges	Decision makers
1	Programme Outline (PO)	
2	Programme Acceptance (PA)	
3	Programme End-results (PE)	
4	Programme Regulation (PR)	

SECTION 3: Major sub barriers
A. Please indicate: 1: Strongly disagree, 2: Disagree, 3: Neutral, 4: Agree, 5: Strongly agree.

SN	Challenges	1	2	3	4	5
1	Absence of a strong, unified stakeholder push for micro-credentials					
2	Absence of a clear definition that can be easily understood without any specific expertise or discipline background					
3	Absence of a recognised volume of learning					
4	Pre-designed learning curriculum that are difficult to divide into smaller sections					
5	Lack of acceptance by one professional or occupational group across multiple jurisdictions					
6	Lack of acceptance by employers					
7	Lack of acceptance by individuals					
8	Mismatch between learning hours and results					
9	Lack of detailed identity verification measures					
10	Lack of detailed process assessment methods					
11	Course descriptions that differ, particularly when every comparable courses are compared, such as learning outcomes					
12	Unclear remuneration policies					
13	Lack of trust between different institutions					
14	Problem of accreditation and quality assurance is the problem of security issue of credentials.					

Source(s): Authors' own

About the authors

Rohit Raj is currently pursuing a master degree at Chaoyang University of Technology, Taichung, Taiwan, in the Department of Information Management. His research interests are in the area of text mining, data analysis, bibliometric analysis, technological innovation and patent analysis, total quality management, manufacturing strategy, supply chain management, strategic management and operation management. He has published twelve papers in reputed journals.

Dr Arpit Singh is Lecturer of Information Systems and Analytics at Jindal Global Business School (JGBS) at O. P. Jindal Global University (Institute of Eminence), India. He has rich industrial experience in the investment banking and private equity research sectors. He holds a PhD in industrial and management engineering from the Indian Institute of Technology (IIT) Kanpur, Uttar Pradesh, and an MTech in Operations Research (Gold Medalist) from National Institute of Technology (NIT) Durgapur, West Bengal, and a BTech in Electrical and Electronics Engineering (First Class) from Dr K.N.Modi Foundation, Ghaziabad, Uttar Pradesh, India. Arpit's research interest is in the areas of decision-making under uncertainty, big data analytics, statistical applications in business problems and adoption of new technologies in the business domain. His work on these research areas has been published in multiple international peer reviewed (ABDC/ABS ranked) journals and top international conferences on advanced computing and intelligent engineering including *Safety Science* (Elsevier), *International Journal of Quality and Reliability Management* (Emerald), and *Behavior and Information Technology* (Taylor & Francis).

Vimal Kumar is Assistant Professor at Chaoyang University of Technology, Taichung, Taiwan (ROC) in the Department of Information Management. He completed his postdoctoral research at Chaoyang University of Technology, Taichung, Taiwan (ROC) in the Department of Business Administration in the domain of technological innovation and patent analysis. He has served as Assistant Professor under TEQIP III, an initiative of MHRD, Govt. of India at AEC Guwahati in the Department of Industrial and Production Engineering. Prior to joining AEC, he served as Assistant Professor at MANIT, Bhopal, in the Department of Management Studies and also served as Visiting Faculty at IMT Nagpur. He obtained his PhD in the domain of TQM and manufacturing strategy in the year 2017 and masters in supply chain management from the Department of Industrial and Management Engineering, IIT Kanpur in the year 2012. He graduated (BTech) in Manufacturing Technology from JSS Academy of Technical Education Noida, in the year 2010. He has published 81 articles in reputable international journals, 9 book chapters, and presented 25 papers at international conferences. His research paper entitled *"Time Table Scheduling for Educational Sector on an*

E-Governance Platform: A Solution from an Analytics Company” has been selected for best paper award at the International Conference on Industrial Engineering and Operations Management (IEOM) held in Bandung, Indonesia, March 6–8, 2018. He was also invited to serve as session chair of session on “Energy Related Awareness” held on 19th September, 2018 at iCAST 2018, IEEE International Conference on Awareness Science and Technology and “Lean Six Sigma” at the International Conference on Industrial Engineering and Operations Management (IEOM-2018) at Bandung, Indonesia, and “Quality Control and Management” at the International Conference on Industrial Engineering and Operations Management (IEOM-2016) at Kuala Lumpur, Malaysia. He has been appointed as an editorial board member in the *IEEE-TEMS Journal* from 1 January 2022 to 31 December 2024. He is a contributing author in international journals including *Journal of Cleaner Production*, *Journal of Informetrics*, *Technology in Society*, *CLSCN*, *Supply Chain Management: An International Journal*, *IJOA*, *JEEE*, *BSE*, *TFSC*, *JKM*, *CSREM*, *IJPPM*, *IJQRM*, *IJPMB*, *IJPQM*, *IJBIS*, *AJOR*, *The TQM Journal*, and *Benchmarking: An International Journal*, etc. and also a guest reviewer of a reputable journal like *IEEE-TEMS*, *JOI*, *IJPPM*, *IJQRM*, *TQM and Business Excellence*, *The TQM Journal*, *Benchmarking: An International Journal*, *Journal of Asia Business Studies*, and *JSIT*. Vimal Kumar is the corresponding author and can be contacted at: vimaljss91@gmail.com

Pratima Verma is currently working as Assistant Professor in the Department of Strategic Management at Indian Institute of Management Kozhikode, India, and associated with Chaoyang University of Technology, Taichung, Taiwan. She has worked as Assistant Professor in Strategic Management area at Indian Institute of Management (IIM) Bodh Gaya, India. Prior to joining IIM, she was postdoctoral fellow in the Department of Management Studies at IIT Madras, India. She obtained her PhD from IIT Kanpur where she worked in the domain of strategic management and horizontal strategy in the Department of Industrial and Management Engineering, in the year 2017. She received her MBA in finance and human resource management from BBDNITM, Uttar Pradesh, and Technical University- Lucknow, India, in the year 2011. She completed her graduation (B.Tech) in Information Technology in the year 2009 from BBNITM, Lucknow. She has one year of experience in teaching. She also awarded JRF/SRF in the area of human resource management. She has published 48 articles in reputable international journals, nine book chapters, and presented fifteen papers at international conferences. Her research paper entitled “*Time Table Scheduling for Educational Sector on an E-Governance Platform: A Solution from an Analytics Company*” has been selected for best paper award in the International Conference on Industrial Engineering and Operations Management (IEOM) held in Bandung, Indonesia, March 6–8, 2018. She was invited to serve as session chair for Human Factors and Ergonomics Track at the International Conference on Industrial Engineering and Operations Management at Kuala Lumpur, Malaysia. She is a contributing author in International journals including *Journal of Cleaner Production*, *Supply Chain Management: An International Journal*, *CLSCN*, *IJOA*, *JEEE*, *IJPPM*, *IJQRM*, *The TQM Journal*, *IJPMB*, *IJISE*, *IJBIS*, and *Benchmarking: An International Journal*, etc.

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